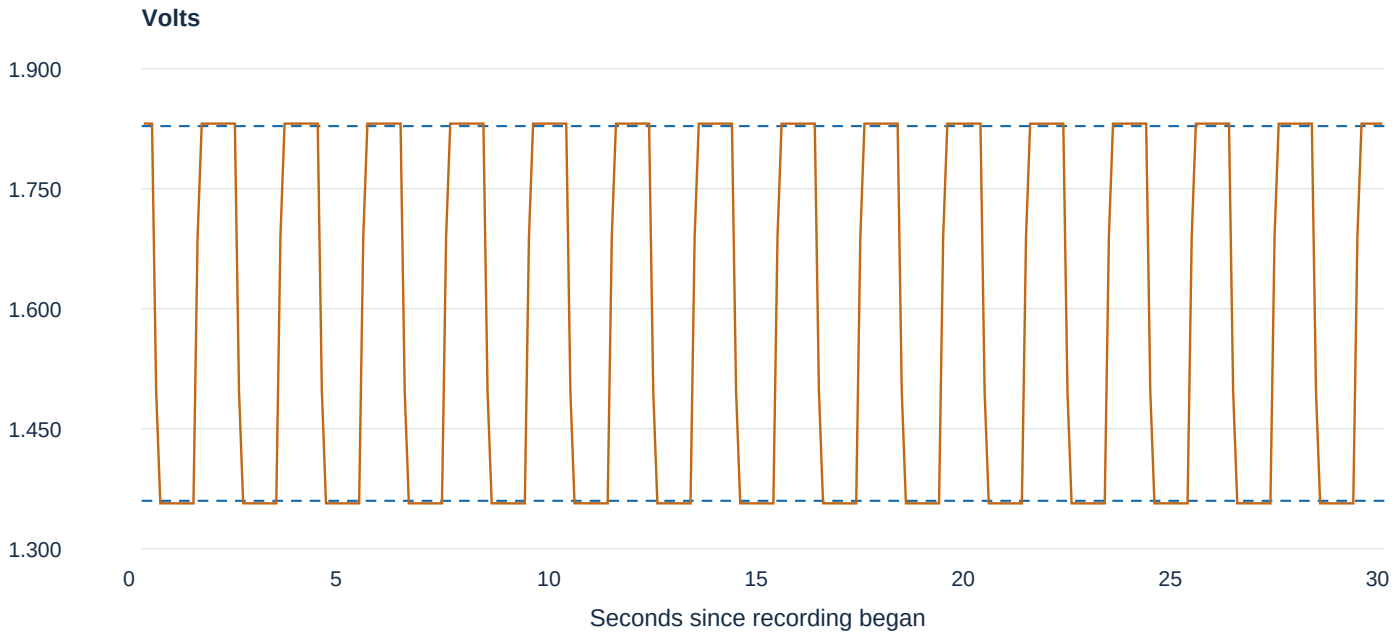
DIGITIZER TEST REPORT

Square-wave recording

Rigol external source | 0.5 Hz, 50% duty cycle | ADS127 at 1,024,000 samples/s

PASS - acquisition and retained SD readback checks

29.999 seconds recorded; no reported receive errors or ring-buffer overflow.
92,158,293 sample bytes; one file changeover; all retained checksums matched.



Recorded levels compared with scope readings

Level	Scope (V)	ADC mean (V)	ADC - scope (mV)	Difference
Vbase	1.3598	1.356657	-3.143	-0.231%
Vtop	1.8282	1.831279	+3.079	+0.168%

ADC swing: 474.621 mV. Scope swing: 468.400 mV. ADC swing is 1.33% larger.

Worst plateau-average difference: 3.182 mV (0.68% of scope swing).

How to read this graph

Orange points average 100 ms of ADC samples read back from the SD card.

Blue lines are the two scope readings entered by the operator, not a simultaneous trace.

Voltage comparisons omit two bins near each detected edge and recording boundary.

The one-second level timing passed. The functional voltage limit was 10% of scope swing.

Differences can include both instruments. ADC conversion assumes a 2.5 V reference.

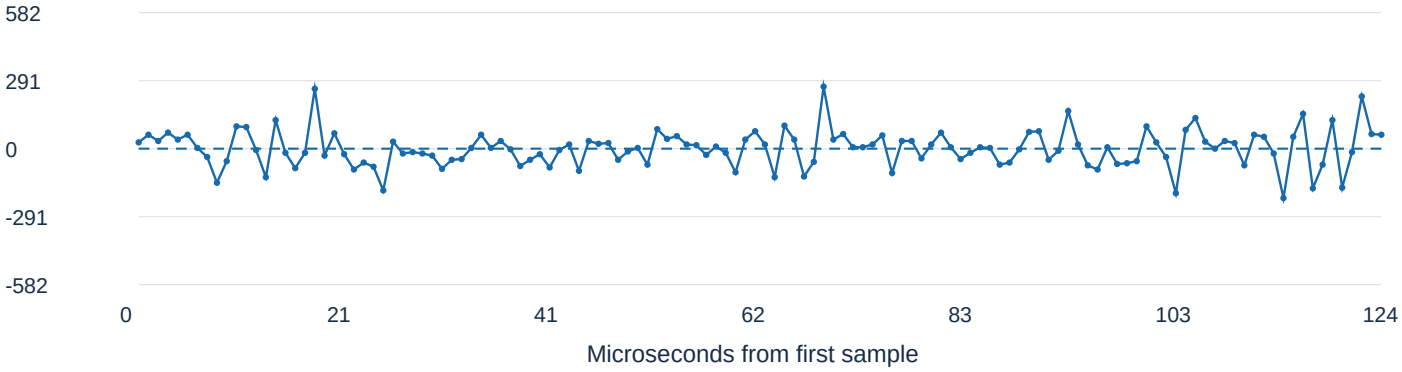
Source files: captures/hardware-smoke/20261030-120712/result.json and p4.txt

Individual-sample fluctuations

Consecutive raw ADC samples from settled sections of the SD recording

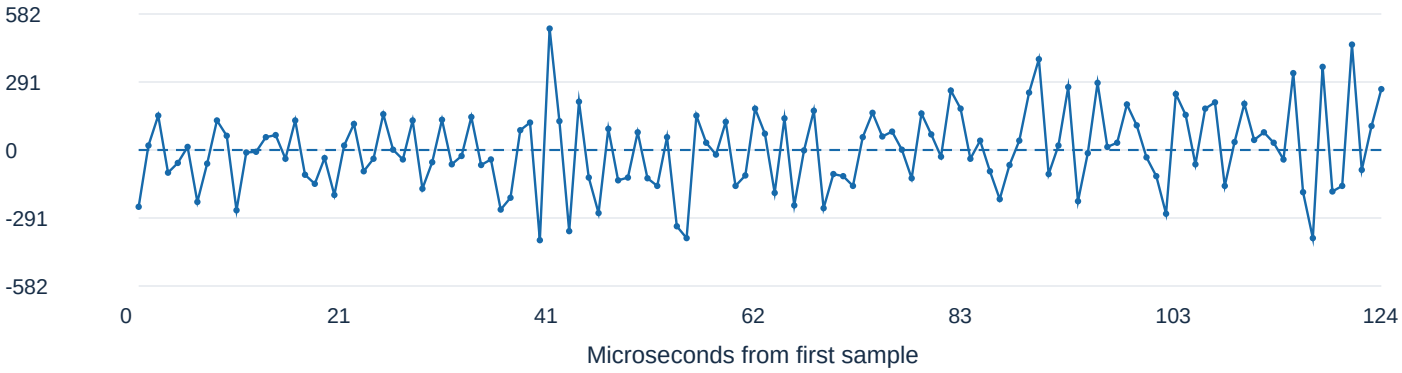
Vbase / low: window starts at 0.550 seconds

Deviation from trace mean (uV)



Vtop / high: window starts at 1.550 seconds

Deviation from trace mean (uV)



Each panel shows 128 samples (124 microseconds), with its own mean subtracted.
Both use the same vertical scale. These short examples do not include every extreme.

Statistics across all selected settled samples	Vbase / low	Vtop / high
Samples analyzed	9,216,000	8,908,800
Individual-sample std. dev. (uV)	172.360	179.583
Individual-sample variance (uV squared)	29707.868	32249.878
Minimum voltage (V)	1.355264	1.829180
Maximum voltage (V)	1.358148	1.836050
Peak-to-peak spread (uV)	2884.865	6869.435
100 ms average std. dev. (uV)	30.459	33.615

Individual samples fluctuate about 172-180 uV (standard deviation); averaging hides much of it.
The high-level maximum of 1.836050 V occurred in the bin starting at 29.6 s, outside these traces.
Statistics include source fluctuation, input-circuit effects and drift; they do not isolate ADC noise.